

General Knowledge

Static GK

STUDY NOTES

Important Scientific Inventions



Important Scientific Inventions

Some famous scientists and their most remarkable inventions and discoveries Invention is something you create by experimentation, whereas discovery is finding out what already exists

Inventions and Discoveries

Invention	Inventor	Year	Country	Description
Electric Iron	H.W. Seeley	1882	U.S.A	An electric appliance used to press clothes and remove wrinkles.
Electric Motor	Moritz Jacobi	1834	Russia	Converts electrical energy into mechanical energy.
Evolution, the theory of	Charles Darwin	1858	England	The theory explains the process of natural selection.
Film sound	Dr. Le de Forest	1923	U.S.A	The technology is used to add audio to motion pictures.
Glider	Sir George Cayley	1853	England	A type of aircraft that does not have an engine.
Insulin	Sir Frederick Banting	1923	Canada	A hormone used to treat diabetes.
Safety Match	J.E. Lundstrom	1855	Sweden	A match designed to ignite only when struck against a specially prepared surface.
Motor car, Petrol	Karl Benz	1885	Germany	One of the first automobiles powered by an internal combustion engine.
Radium	Marie & Pierre Curie	1898	France	A radioactive element used in cancer treatment.

Rubber (vulcanised)	Charles Goodyear	1841	U.S.A	Processed rubber that is more durable and elastic.
Safety Lamp	Sir Humphry Davy	1816	England	A lamp designed for safe use in explosive atmospheres.
Telescope	Hans Lippershey	1608	Netherlands	An optical instrument used to observe distant objects.
Television	John Logic Baird	1926	Scotland	A device for transmitting visual media.
Thermometer	Galileo	1593	Italy	A device used to measure temperature.
Valve, Radio	Sir J.A Fleming	1904	Britain	An electronic component is used in radio receivers.
Printing Press	Johannes Gutenberg	1440	Germany	A machine for mass-producing printed materials.
Pocket Watch	Peter Henlein	1510	Germany	A small, portable timepiece.
Microscope	Zacharias Janssen	1590	Netherlands	An instrument for viewing small objects.
Logarithms, Napier Bones & decimal point	John Napier	1590s	Scotland	Mathematical tools and notations.
Automatic Calculator	Wilhelm Schickard	1623	Germany	An early mechanical calculator.
Adding Machine	Blaise Pascal	1642	France	A mechanical device for basic arithmetic.
Barometer	Evangelista Torricelli	1643	Italy	Measures atmospheric pressure.
Air Pump	Otto Von Guericke	1650	Germany	A device that removes air from a sealed volume.
Bacteria	Antonie Van Leeuwenhoek	1665	Netherlands	Discovered using a microscope.

Pendulum Clock	Christian Huygens	1657	Netherlands	A highly accurate timekeeping device.
Gravity & Reflecting Telescope	Isaac Newton	1668	England	A telescope design using mirrors.
Clarinet	Johann Christoph Denner	1690	Germany	A type of woodwind musical instrument.
Steam Engine	Thomas Savery	1698	UK	An engine that uses steam to generate power.
Piano	Bartolomeo Cristofori	1700	Italy	A large keyboard musical instrument.
Centigrade Scale	Anders Celsius	1742	Sweden	A temperature scale where 0 is the freezing point and 100 is the boiling point of water.
Electroscope	Jean Nolle	1748	France	An early scientific instrument for detecting electric charge.
Lightning Conductor	Benjamin Franklin	1752	USA	A device for protecting buildings from lightning.
Hydrogen	Henry Cavendish	1766	England	Discovered as a chemical element.
Chlorine	Karl Wilhelm Scheele	1774	Sweden	Discovered as a chemical element used in sanitation.
Ship (Steam)	J.C Perrier	1775	France	An early steam-powered vessel.
Oxygen	Antoine Laurent Lavoisier	1775	France	Discovered as a chemical element.
Submarine	David Bushnell	1776	USA	An underwater vessel.

Hot Air Balloon	Josef & Etienne Montgolfier	1783	France	Used for aerial travel using heated air.
Tungsten	Juan José Elhuyar Lubize & Fausto de Elhuyar	1783	Spain	Discovered as a chemical element.
Bifocal Lens	Benjamin Franklin	1784	USA	Eyeglasses with lenses of two different optical powers.
Parachute	Jean Pierre Blanchard	1785	France	A device used to slow descent from high altitudes.
Steam Boat	John Fitch	1786	USA	An early steam-powered boat.
Guillotine	Dr. Joseph-Ignace Guillotin	1790s	France	A device for executing by beheading.
Watch (Self Winding)	Abraham-Louis Breguet	1791	France	A watch that winds itself using the motion of the wearer's wrist.
Milling machine & Cotton gin	Eli Whitney	1793	USA	Machines for processing cotton and machining.
Smallpox Vaccine	Edward Jenner	1796	England	First successful vaccine for smallpox.
Electric Battery	Volta	1800	Italian	A device that stores electrical energy.
Locomotive	Richard Trevithick	1804	England	An early steam-powered train.
Stethoscope	Rene Laennec	1816	France	A medical instrument used for listening to the internal sounds of the body.

Dental Plate	Anthony A. Plantson	1817	USA	A removable dental prosthesis.
Cadmium	Friedrich Stromeyer	1817	Germany	Discovered as a chemical element.
Cement	Joseph Aspdin	1824	England	Invented the process for making Portland cement.
Electromagnet	William Sturgeon	1824	England	An early form of electromagnet using iron.
Photography (on metal)	Joseph Nicéphore Niepce	1826	France	Early photographic process using metal plates.
Braille	Louis Braille	1829	France	A system of raised dots is used by blind individuals to read.
Electromagnetic Induction	Michael Faraday	1831	UK	The process by which a magnetic field can create an electric current.
Electric Generator	Michael Faraday	1831	USA	A device that converts mechanical energy into electrical energy.
Revolver	Samuel Colt	1835	USA	A type of repeating firearm.
Photography (on paper)	W. H. Fox Talbot	1835	England	Introduced the negative-positive process, allowing multiple copies of an image.
Telegraph	William Cook & Charles Wheatstone	1837	England	Revolutionised long-distance communication by transmitting electrical signals.
Telegraph code	Samuel Morse	1837	USA	Developed Morse code, a system of dots and dashes used to send messages.

Ozone	Christian Schonbein	1839	Germany	Discovered ozone, an allotrope of oxygen, crucial for absorbing UV radiation.
Typewriter	Peter Mitterhofer	1841	Austria	Early mechanical writing devices, simplified document creation and correspondence.
Transformer (Induction coil)	William Stanley Jr.	1842	USA	Key component in electrical circuits, transferring electrical energy between circuits.
Saxophone	Adolphe Sax	1846	Belgium	Woodwind instruments are widely used in jazz, classical, and contemporary music.
Sewing Machine	Elias Howe	1846	USA	Mechanized sewing, vastly improving the efficiency and consistency of garment production.
Motorcycle	Edward Butler	1848	England	One of the earliest forms of motorized transportation, paving the way for modern motorcycles.
Bullet	Claude Minie	1849	France	Developed the Minié ball, a conical bullet that improved the accuracy and range of firearms.
Safety Pin	William Hunt	1849	USA	Simple fastening device, widely used for securing fabric and clothing.
Burglar Alarm	Edwin T. Holmes	1851	USA	Early security system designed to alert property owners of unauthorized entry.

Refrigerator	James Harrison	1851	Australia	Early mechanical refrigeration system, crucial for food preservation.
Elevator	Elisha G. Otis	1852	USA	Introduced a safety mechanism for elevators, making high-rise buildings feasible.
Steam-Powered Airship	Henri Giffard	1852	France	First successful powered airship, controlled flight over a distance.
Light Bulb	Heinrich Goebel	1854	Germany	Created an early version of the incandescent light bulb, leading to electric lighting.
Generator	Piciontti	1860	Italy	Early device for converting mechanical energy into electrical energy.
Machine Gun	Richard Gaffing	1861	USA	Early rapid-fire weapons, significantly impacted military tactics and technology.
Typewriter	Peter Mitterhofer	1864	Austria	Improved version of the early typewriter, enhancing writing efficiency.
Antiseptic	Dr. Joseph Lister	1867	England	Introduced antiseptic surgical methods, reducing infections and improving patient outcomes.
Dynamite	Alfred B. Nobel	1867	Sweden	A powerful explosive used in construction, mining, and demolition.

Helium	William Ramsay	1868	Great Britain	Discovered the noble gas used in balloons and as a coolant in various applications.
Vacuum Cleaner	Ives McGuffey	1869	USA	An early version of a household device for cleaning floors.
Electric Motor (DC)	Zenobe Gramme	1873	Belgium	Developed the direct current (DC) motor, revolutionising mechanical power applications.
Barbed Wire	Joseph F. Glidden	1873	USA	Improved fencing material, widely used in agriculture for livestock control.
Telephone	Alexander Graham Bell	1874	Canada	Invented a device that revolutionised communication by transmitting voice over distances.
Microphone	Alexander Graham Bell	1876	USA	Enhanced sound transmission and recording technology.
Carburetor	Gottlieb Daimler	1876	Germany	A device for mixing air with fuel in internal combustion engines.
Phonograph	Thomas Edison	1877	USA	Early device for recording and playing back sound.
Welder (Electric Welding)	Elisha Thompson	1877	USA	Pioneered electric welding techniques, essential for metalworking industries.
Gramophone	Thomas Edison	1878	USA	Improved phonograph, making recorded music more accessible.

Cash Register	James Ritty	1879	USA	Invented to help business owners keep track of sales and cash flow.
Arc Lamp	C. F. Brush	1879	Ohio, USA	An early form of electric lighting that uses an electric arc to produce light.
Electric Lamp	Thomas Alva Edison	1879	USA	Developed the practical incandescent light bulb.
Electromagnetic Theory	Heinrich Rudolph Hertz	1880	Germany	Proved the existence of electromagnetic waves, forming the basis for radio technology.
AC Motor and Transformer	Nikola Tesla	1880s	USA	Key contributions to the development of alternating current (AC) electrical systems.
Electric Fan	Wheeler	1882	USA	Early invention for cooling and ventilation using electric power.
Fountain Pen	Lewis Edson Waterman	1884	USA	Improved the design of fountain pens, making them more reliable for writing.
Fluorine	Ferdinand Frederick Henri Moissan	1886	France	Isolated the element fluorine, used in various chemical applications.
Ball Point Pen	John Loud	1888	USA	Developed an early version of the modern ballpoint pen.
Bicycle Tyres	John Boyd Dunlop	1888	Scotland	Invented pneumatic tires, significantly improving bicycle comfort and performance.

Photography (on film)	John Carbutt	1888	USA	Innovated the use of photographic film, leading to modern photography.
Time Recorder	Harlow Bundy	1890	USA	An early device used for tracking employee work hours.
Animation	Emile Reynaud	1892	France	Pioneer of animated films created the first animated projections.
Cinema	Lumiere brothers	1894	France	Invented the cinematograph, an early motion-picture camera and projector.
Argon	William Ramsay & Baron Rayleigh	1894	UK	Discovered the noble gas argon, used in lighting and welding.
Radio	Guglielmo Marconi	1894	Italy	Developed the first practical wireless telegraphy system.
Ship (Turbine)	Charles Parsons	1894	England	Invented the steam turbine engine, revolutionizing marine propulsion.
Diesel Engine	Rudolf Diesel	1895	Germany	Invented the diesel engine, widely used in heavy machinery and transportation.
X-ray	Wilhelm Conrad Roentgen	1895	Germany	Discovered X-rays, transforming medical diagnostics.
Electric Stove/Cooker	William S. Hadaway	1896	USA	Early electric cooking appliance, improving kitchen convenience.

Aspirin	Dr. Felix Hoffman	1899	Germany	Developed aspirin, a widely used pain reliever and anti-inflammatory drug.
Cellophane	I.E. Brandenberger	1900	Switzerland	Invented cellophane, used for packaging and wrapping.
Tractor (Caterpillar)	Benjamin Holt	1900	USA	Developed the first continuous track tractors, enhancing agricultural efficiency.
Disc Brake	Dr. F. Lanchester	1902	England	Invented the disc brake system, improving vehicle safety and performance.
Windshield Wipers	Mary Anderson	1903	USA	Invented the first practical windshield wiper, enhancing driver visibility.
Airplanes	Wilbur and Orville Wright	1903	USA	Achieved the first powered, controlled flight with their aircraft.
Silicones	F.S Kipping	1904	England	Developed silicone compounds, used in various industrial and medical applications.
Bakelite	H. Backland	1907	Belgium/USA	Invented Bakelite, the first synthetic plastic, revolutionizing materials science.
Gyrocompass	Elmer A. Sperry	1908	USA	Invented the gyrocompass, essential for navigation in ships and aircraft.

Vitamin A	Elmer V. McCollum & M. Davis	1913	USA	Discovered vitamin A, essential for vision and immune function.
Air Conditioner	Willis Carrier	1914	USA	Developed the first modern air conditioning system, enhancing indoor comfort.
Neon Lamp	Georges Claude	1915	France	Invented the neon lamp, widely used in advertising and signage.
Vitamin B	Elmer V. McCollum	1916	USA	Discovered vitamin B complex, crucial for energy metabolism.
Motor Scooter	Greville Bradshaw	1919	England	Developed an early version of the motor scooter, making personal transportation more accessible.
Vitamin C	Albert Szent-Györgyi & Charles Glen King	1920	USA	Discovered vitamin C, essential for collagen synthesis and antioxidant protection.
Vitamin D	Edward Mellanby	1920	USA	Discovered vitamin D, crucial for calcium absorption and bone health.
Band Aid	Earle Dickson	1920	USA	Invented the adhesive bandage, providing convenient wound care.
Crescograph	Jagadish Chandra Bose	1920s	India	Invented the crescograph, used to measure plant growth responses.
Insulin	Frederick Banting &	1921	Canada	Discovered insulin, a hormone essential for the treatment of diabetes.

	Charles H. Best			
Radar	Dr. Albert H. Taylor & Leo C. Young	1922	USA	Developed radar technology, crucial for detecting objects using radio waves.
Vitamin E	Herbert McLean Evans	1922	USA	Discovered vitamin E, important for its antioxidant properties.
Film (Musical)	Warner Bros	1923	USA	Pioneered the creation of musical films, combining song and dialogue in cinema.
Adhesive Tape	Richard G. Drew	1923	USA	Invented adhesive tape, widely used for sealing and packaging.
Loud Speaker	Chester W. Rice and Edward W. Kellogg	1924	USA	convert electrical audio signals into sound waves
Film (Talking)	Warner Bros	1926	USA	Introduced synchronized sound to motion pictures, revolutionizing the film industry.
Rocket Engine	Robert H. Goddard	1926	USA	Developed the first liquid-fueled rocket, laying the foundation for modern rocketry.
Penicillin	Alexander Fleming	1928	England	Discovered the first antibiotic, which has saved countless lives by treating bacterial infections.

Xerox Machine	Chester Carlson	1928	USA	Invented the photocopier, revolutionizing document reproduction.
Car Radio	William Lear & Elmer Wavering	1929	USA	Developed the first car radio, enhancing in-vehicle entertainment.
Vitamin K	Henrik Dam & Edward Adelbert Doisy	1929	Denmark & USA	Discovered the vitamin essential for blood clotting and bone health.
Richter Scale	Charles Richter	1935	USA	Created a logarithmic scale to quantify the magnitude of earthquakes.
Jet Engine	Hans Von Ohain	1936	Germany	Developed the first operational jet engine, transforming aviation.
Microscope (Electron)	Vladimir Kosme Sworykin	1939	Russia	Invented the electron microscope, greatly increasing the magnification and resolution of microscopic images.
Helicopter	Igor Sikorsky	1939	Russia	Developed the first successful helicopter, revolutionizing vertical flight.
Atom Bomb	Julius Robert Oppenheimer	1945	USA	Led the development of the atomic bomb during the Manhattan Project.
Microwave Oven	Percy Spencer	1947	USA	Invented the microwave oven, transforming food preparation.

Transistors	John Bardeen, William Shockley & Walter Brattain	1948	USA	Developed the first transistors, critical components in modern electronic devices.
Pacemaker	Dr. Paul Zoll	1952	USA	Developed the first successful cardiac pacemaker to regulate heartbeats.
Electronic Calculator	IBM	1954	USA	Introduced the first practical electronic calculator, enhancing computational efficiency.
Hovercraft	Christopher Cockerell	1959	England	Invented the hovercraft, capable of travelling over various surfaces by hovering on a cushion of air.
Laser	Dr. Charles H. Townes	1960	USA	Developed the first laser, a device that emits a narrow beam of coherent light.
Soft Contact Lenses	Otto Wichterle	1961	Czech Republic	Invented the first soft contact lenses, improving comfort for vision correction.
Synthesizer	Dr. Robert Arthur Moog	1964	USA	Created the first commercial synthesizer, revolutionizing electronic music.
Bicycle	Kirkpatrick Macmillan	1839	Britain	Invented an early version of the modern bicycle with pedals.
Celluloid	Alexander Parkes	1861	Britain	Developed the first synthetic plastic, used in film production and other applications.

Chloroform	E. Soberran	1831	France	Discovered chloroform, later used as an anaesthetic in medical surgeries.
Cine Camera	Wm. Friese-Greene	1889	Britain	Invented an early motion picture camera, contributing to the development of cinema.
Circulation of Blood	William Harvey	1628	England	Discovered the circulation of blood in the human body, transforming medical understanding.
Clock (Mechanical)	Hsing and Ling-Tsan	1725	China	Developed one of the earliest mechanical clocks, contributing to timekeeping advancements.
Diesel Engine	Rudolf Diesel	1892	Germany	Invented the diesel engine, widely used for its efficiency in heavy machinery and transportation.